Exercise 3

Problem 1 (Gradient Descent for Least-square Problem):
Consider the problem of solving the linear system

$$\mathbf{X}\mathbf{w} = \mathbf{y}$$

where $\mathbf{X} = (\mathbf{x}_i^\top)_{i=1}^n \in \mathbb{R}^{n\times d}$ is the data matrix, $\mathbf{w} \in \mathbb{R}^d$ the parameter vector of the model, and $\mathbf{y} \in \mathbb{R}^n$ the target. We assume that there exists a unique solution provided by the mean-squared loss

$$\mathbf{w}_* = \arg\min_{\mathbf{w}\in\mathbb{R}^d} \left(\frac{1}{2}\|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 =: \ell(\mathbf{w}) \right).$$

We consider the gradient descent method

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \nabla \ell(\mathbf{w}_t) \tag{1}$$

with a fixed step size η.

a) Calculate the gradient $\nabla\ell(\mathbf{w})$ and use it to re-write the gradient descent iterates of Eq. (1).

b) Calculate the Hessian $\nabla^2\ell(\mathbf{w})$ and argue whether or not $\ell(\mathbf{w})$ is convex.

a) $\nabla \ell(w) = X^\top (Xw - y)$ using vector calculus

b) $\nabla^2 \ell(w) = X^\top X$, which is positive semi-definite. (why?)

A positive semi-definite Hessian implies that $\ell(w)$ is convex.
→ ℓ only has positive curvature.

c) Use the following facts.

- Remember that $\mathbf{y} = \mathbf{X}\mathbf{w}_*$; → does not hold in general of course! Only if $\ell(w_*) = 0$
- Recall the definition of the operator norm of a matrix:

$$\|\mathbf{A}\|_{\text{op}} = \sup\left\{ \frac{\|\mathbf{A}\mathbf{x}\|_2}{\|\mathbf{x}\|_2} : \mathbf{x} \in \mathbb{R}^d \text{ with } \mathbf{x} \neq 0 \right\} = \sqrt{\lambda_{\max}(\mathbf{A}^T\mathbf{A})},$$

  where $\lambda_{\max}(\mathbf{A}^T\mathbf{A})$ denotes the largest eigenvalue of $\mathbf{A}^T\mathbf{A}$.
- for any matrix $\mathbf{A} \in \mathbb{R}^{d\times d}$ and vector $\mathbf{v} \in \mathbb{R}^d$, we have $\|\mathbf{A}\mathbf{v}\|_2 \leq \|\mathbf{A}\|_{\text{op}} \cdot \|\mathbf{v}\|_2$, where $\|\mathbf{A}\|_{\text{op}}$ denotes the operator norm of the matrix $\mathbf{A}$;

Show that the following bound on the distance to the optimizer holds

$$\|\mathbf{w}_{t+1} - \mathbf{w}_*\|_2 \leq \|\mathbf{I} - \eta\mathbf{X}^\top\mathbf{X}\|_{\text{op}}^t \cdot \|\mathbf{w}_1 - \mathbf{w}_*\|_2 \tag{5}$$

Let's just plug in the gradient into the optimizer update:

$$w_{k+1} = w_k - \eta \nabla \ell(w_k)$$

$$= w_k - \eta X^\top (Xw_k - y) = w_k - \eta X^\top (Xw_k - Xw_*)$$

$$w_{k+1} - w_* = (w_k - w_*) - \eta X^\top X (w_k - w_*) = (I - \eta X^\top X)(w_k - w_*)$$

Applying this recursively:

$$w_{k+1} - w_* = (I - \eta X^T X)^k (w_1 - w_*)$$

Taking the norm on both sides:

$$\|w_{k+1} - w_*\| \leq \|(I - \eta X^T X)^k (w_1 - w_*)\|$$

$$\leq \|(I - \eta X^T X)^k\|_{op} \cdot \|w_1 - w_*\|$$

$$\leq \|I - \eta X^T X\|_{op}^k \cdot \|w_1 - w_*\|$$

d) Show that, if $\eta < \frac{1}{\|\mathbf{X}^\top \mathbf{X}\|_{\text{op}}}$, i.e. if $\|\eta \mathbf{X}^\top \mathbf{X}\|_{\text{op}} < 1$, then the Gradient Descent converge, i.e. $\|\mathbf{w}_{t+1} - \mathbf{w}_*\|_2 \to 0$ as $t \to \infty$.

e) One, obviously cheaper, alternative is to only compute the update (indexed by k) based on the gradient of one specific datapoint $\mathbf{x}_i$. This is the updated of *stochastic* gradient descent (SGD), which is arguably the most widely used optimizer in ML:

$$\mathbf{w}_{k+1} = \mathbf{w}_k - \eta \nabla \ell_i(\mathbf{w}_k), \quad i \in \{1, \ldots, n\};\ \eta > 0.$$

where we denote $\ell_i(\mathbf{w}) := \frac{n}{2}(\mathbf{x}_i^\top \mathbf{w} - y_i)^2$ for each $i = 1, ..., n$. In each iteration k, a datapoint $\mathbf{x}_i$ is chosen uniformly at random such that $\mathbb{E}[\nabla \ell_i(\mathbf{w}_k)] = \nabla \ell(\mathbf{w}_k)$.

Show that, given a constant positive step size $\eta \leq \frac{1}{2\|\mathbf{X}^\top \mathbf{X}\|_{\text{op}}}$, we have

$$\mathbb{E}\left[\|\mathbf{w}_{k+1} - \mathbf{w}_*\|_2^2\right] \geq \eta^2 \mathbb{E}\left[\|\nabla \ell_i(\mathbf{w}_k)\|_2^2\right] > 0$$

In other words, SGD does not converge to the critical point $\mathbf{w}_*$ on average.

d) we need: $\|I - \eta X^T X\|_{op} < 1$ for $\|w_{k+1} - w_*\|_2 \to 0$ as $k \to \infty$

$$\|I - \eta X^T X\|_{op} = |\lambda_{max}(I - \eta X^T X)| = |1 - \lambda_{max}(\eta X^T X)|$$

eigenvalue-decomp.: if $X^T X = V \Lambda V^T$ then $I - \eta X^T X = VV^T - \eta V \Lambda V^T = V(I - \eta\Lambda)V^T$

$$= |1 - \eta \lambda_{max}(X^T X)| < 1$$

if $\eta \lambda_{max}(X^T X) < 1$, this ineq. holds! But actually we just need: $\eta \lambda_{max}(X^T X) < 2$.

$$(1 - \eta \|X^T X\|_{op}) < 1 \quad \eta > 0$$

$$-(1 - \eta \|X^T X\|_{op}) < 1$$

$$\eta < \frac{2}{\|X^T X\|_{op}}$$

e) Let's proceed the same way as in part c):
(except now we need to take the expectation)

$$\|w_{k+1} - w_*\|_2^2 = \|w_k - \eta \nabla \ell_i(w_k) - w_*\|^2$$

$$= \|w_k - w_* - \eta \nabla \ell_i(w_k)\|_2^2$$

$$= \|w_k - w_*\|^2 + \eta^2 \|\nabla \ell_i(w_k)\|_2^2 - 2\eta \langle w_k - w_*, \nabla \ell_i(w_k) \rangle$$

Taking expectation on both sides:

$$\mathbb{E}_k \|w_{k+1} - w_*\|_2^2 = \|w_k - w_*\|^2 + \eta^2 \mathbb{E}_k \|\nabla \ell_i(w_k)\|_2^2 - 2\eta \langle w_k - w_*, \mathbb{E}_k \nabla \ell_i(w_k) \rangle$$

($\nabla \ell(w_k)$ = $\mathbb{E}_k \nabla \ell_i(w_k)$)

$$\overset{\text{Cauchy-Schwarz}}{\geq} \|w_k - w_*\|^2 + \eta^2 \mathbb{E}_k \|\nabla \ell_i(w_k)\|_2^2 - 2\eta \|w_k - w_*\| \cdot \|\nabla \ell(w_k)\|$$

$$= \|w_k - w_*\|^2 + \eta^2 \mathbb{E}_k \|\nabla \ell_i(w_k)\|_2^2 - 2\eta \|w_k - w_*\| \cdot \|\nabla \ell(w_k) - \underbrace{\nabla \ell(w_*)}_{=0}\|$$

$$\overset{\text{Lipschitz smoothness}}{\geq} \|w_k - w_*\|^2 + \eta^2 \mathbb{E}_k \|\nabla \ell_i(w_k)\|_2^2 - 2\eta \cdot \underbrace{\|X^T X\|_{op}}_{\leq L = \|X^T X\|_{op}} \cdot \|w_k - w_*\|^2$$

$$= \underbrace{(1 - 2\eta \|X^T X\|_{op})}_{\text{for } \eta < \frac{1}{2\|X^T X\|_{op}}} \cdot \|w_k - w_*\|^2 + \eta^2 \mathbb{E}_k \|\nabla \ell_i(w_k)\|^2$$

$\eta \in \frac{1}{2\|X^T X\|_{op}}$

this term is non-negative

$$\geq \eta^2 \mathbb{E}_k \|\nabla \ell_i(w_k)\|^2 > 0 \quad \square$$

Problem 2 (Polyak's momentum on a quadratic objective):
Consider Polyak's momentum method

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \nabla f(\mathbf{x}_t) + \beta(\mathbf{x}_t - \mathbf{x}_{t-1}), \quad \eta > 0, \quad 0 \le \beta < 1.$$

We study its behavior on a quadratic objective

$$f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^\top \mathbf{H}\mathbf{x}, \qquad H \succeq 0$$

where $\mathbf{x} \in \mathbb{R}^d$ and $\mathbf{H} \in \mathbb{R}^{d \times d}$ is symmetric.

a) Let $\mathbf{H} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$ be an eigendecomposition of $\mathbf{H}$, and define

$$\mathbf{z}_t = \mathbf{Q}^\top \mathbf{x}_t. \iff \mathbf{x}_t = \mathbf{Q}\mathbf{z}_t$$

Show that the coordinate z_t^i of $\mathbf{z}_t$ evolve independently.

b) In an eigendirection corresponding to eigenvalue λ_i, show that

$$z_{t+1}^i = (1 + \beta - \eta\lambda_i) z_t^i - \beta z_{t-1}^i.$$

Thus, the behavior of momentum on a quadratic can be understood by studying a scalar second-order recurrence.

a) $f(x) = \frac{1}{2} x^T H x \quad H = Q\Lambda Q^T$

$= \frac{1}{2} x^T Q \Lambda Q^T x$

$= \frac{1}{2} z^T \Lambda z = \frac{1}{2} \sum_{i=1}^{d} \lambda_i z_i^2$

$\nabla f(x) = Hx = Q\Lambda Q^T x$

$x_{t+1} = x_t - \eta \nabla f(x_t) + \beta (x_t - x_{t-1})$

$Q z_{t+1} = Q z_t - \eta Q \Lambda z_t + \beta (Q z_t - Q z_{t-1})$

$\iff z_{t+1} = z_t - \eta \Lambda z_t + \beta (z_t - z_{t-1})$

$\downarrow$ element-wise

$z_{t+1}^i = z_t^i - \eta \lambda_i z_t^i + \beta (z_t^i - z_{t-1}^i)$

b) $z_{t+1}^i = (1 + \beta - \eta\lambda_i) z_t^i - \beta z_{t-1}^i$

c) For the remaining part of the exercise, we will drop the subscript and write z_t instead of z_t^i.
For a linear recurrence with constant coefficients, try a solution of the form $z_t = r^t$. Why is such a form particularly convenient under time shifts?
↳ time-shifted terms remain proportional to r^t.
Derive the characteristic equation

$$r^2 - (1 + \beta - \eta\lambda_i)r + \beta = 0.$$

Explain why convergence in this eigendirection requires both characteristic roots to satisfy $|r| < 1$. Under what condition are the roots complex and what does this mean qualitatively for the optimization trajectory?

c) $z_{t+1} = (1+\beta-\eta\lambda_i) z_t - \beta z_{t-1}$

$r^{t+1} = (1+\beta-\eta\lambda_i) r^t - \beta r^{t-1} \quad | : r^{t-1}$

$r^2 = (1+\beta-\eta\lambda_i) r - \beta \iff r^2 - (1+\beta-\eta\lambda_i) r + \beta = 0$

roots: $r_{1/2} = \frac{1+\beta-\eta\lambda_i}{2} \pm \sqrt{\left(\frac{1+\beta-\eta\lambda_i}{2}\right)^2 - \beta}$

→ general solution $z_t = c_1 r_1^t + c_2 r_2^t$

⇒ we need $|r| < 1$ for dampening effect s.t. $z_t \to 0$ which is the global minimum of $f(x)$.

complex, when < 0
⇒ $\beta > \left(\frac{1+\beta-\eta\lambda_i}{2}\right)^2$
→ oscillations of the trajectory.

d) The loss landscape of a neural network is generally nonconvex. Nevertheless, one can locally study its behavior through Hessian eigendirections. Consider again the characteristic equation

$$r^2 - (1 + \beta - \eta\lambda_i)r + \beta = 0.$$

Discuss what happens in the following two cases:

(i) Flat direction: $\lambda = 0$.
Show that in this case

$$r^2 - (1 + \beta)r + \beta.$$

What are the roots and what does this imply about the motion in a perfectly flat direction?

(ii) Negative curvature: $\lambda < 0$.
Let

$$p(r) = r^2 - (1 + \beta - \eta\lambda)r + \beta.$$

Show that $p(1) = \eta\lambda < 0$.
Further argue that there must be a root satisfying $r > 1$. (*Hint*: Note that $p(r) \to +\infty$ as $r \to +\infty$).
What does this say about the stability of momentum close to a saddle point? Why might instability along a direction of negative curvature actually be useful during neural network optimization?

d) (i) if $\lambda = 0$, the $\eta\lambda_i r$ term drops and we have:

$r^2 - (1+\beta) r + \beta = 0$

The roots are $r_{1/2} = \frac{1+\beta}{2} \pm \sqrt{\left(\frac{1+\beta}{2}\right)^2 - \beta} = \frac{1+\beta}{2} \pm \sqrt{\frac{1 - 2\beta + \beta^2}{4}} \Rightarrow \begin{matrix} r_1 = 1 \\ r_2 = \beta \end{matrix}$

$\frac{1-\beta}{2}$

Because one root $r_n = 1$ there is no movement along a perfectly flat direction.

(ii) $p(r) = r^2 - (1+\beta-\eta\lambda)r + \beta$

$p(1) = 1 - (1+\beta-\eta\lambda) + \beta = \eta\lambda < 0$ because $\eta > 0$, $\lambda < 0$.

Since $p(1) < 0$, but $p(r) \to +\infty$ for $r \to \infty$, $p(r)$ must cross the x-axis at some $r > 1$, implying a root $r > 1$.

$r > 1$ implies divergence, so at saddle points $\exists \lambda < 0$ the update is unstable.

This can be desirable during NN optimization because it prevents the optimizer being stuck at a saddle point.